

Electoral Judicialization and Municipal Competition in Brazil

Evidence from a Bartik Shift-Share Instrument

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Does Judicial Intervention Shape Electoral Competition?

Research question:

Does pre-election judicial intervention causally affect the competitive dynamics of mayoral races?

The mechanism:

- Candidates face adversarial lawsuits *before* election day
- Lawsuits impose legal costs, reputational exposure, and campaign disruption
- This could deter entry, shift vote shares, and alter voter behavior

The challenge:

Close races *attract* more lawsuits — endogeneity runs in both directions

This Paper

- 5,560 Brazilian municipalities
- 2020 and 2024 mayoral elections
- TSE administrative data
- Identification: **Bartik shift-share IV** ($F \approx 19.6$)

Key Findings

Adversarial judicialization $\uparrow$ significantly raises the blank vote rate (+1.3 pp, $p = 0.018$). Candidate pool contracts (directional). Competition and turnout: precise nulls after tF correction.

Brazil's Electoral Justice System:

- *Tribunal Superior Eleitoral* (TSE) + 27 *Tribunais Regionais Eleitorais* + 2,800+ first-instance zones
- Adjudicates eligibility, campaign finance, and conduct *before* election day
- Among the world's most active electoral court systems
- Rich variation in topic mix across municipalities, states, and cycles

Municipal Setting

- 5,570 simultaneous contests = large cross-section
- Homogeneous office: directly elected mayor
- Uniform electoral rules (CF Art. 29)
- Comparable competitive dynamics across units

Two Election Cycles

2020 (Nov 15) → **shares**
2024 (Oct 6) → **treatment + outcomes**

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Three-tier judicial structure:

- 1 **Zona Eleitoral (ZE)** — first instance; 2,800+ nationwide
- 2 **Tribunal Regional Eleitoral (TRE)** — 27 state appellate courts
- 3 **Tribunal Superior Eleitoral (TSE)** — national apex

Pre-election jurisdiction:

- Candidate registration challenges (*impugnação*)
- Abuse of economic/political power
- Campaign finance violations
- Prohibited propaganda; vote-buying
- Conduct prohibitions (*conduta vedada*)

Election Timeline

- 1 Candidate registration (August)
- 2 Pre-election lawsuits (Aug–Oct)
- 3 **Election Day** (Oct/Nov)
- 4 Post-election proceedings
- 5 Cassation if ruling goes against winner

2020: Election day = November 15

2024: Election day = October 6

We use only cases filed *before* these dates.

Effects on targeted candidates:

- **Candidacy at risk:** court may rule ineligible before election day
- **Reputational cost:** public filing, media coverage, social stigma
- **Resource diversion:** legal defense competes with campaigning
- **Quality signal to voters:** a respondent may be perceived as ethically challenged — or as a victim of political harassment

Who files?

Opposing candidates, parties, *Ministério Público Eleitoral*, any citizen

Treatment Variable

Adversarial competition-relevant cases:

- Mayoral candidate is the *respondent*
- Filed before election day
- Substantive challenge (adversarial class and subject — not administrative)

Measurement:

$\Delta \log(1 + \text{lawsuits}_m)$
from 2020 to 2024

Administrative vs. Adversarial Electoral Litigation

Not all electoral case subjects involve genuine legal challenges to candidates:

Candidacy Registration (RRC): Automatically generated for every registered candidacy. No plaintiff, no adversarial content, no legal risk. Purely administrative.

Coalition Slate Registration (DRAP): Filed by each party before every election; mandatory under Art. 6, Lei 9.504/1997. Universal coverage (>97% of municipalities).

Adversarial challenges: Filed against a specific candidate's conduct, eligibility, or campaign behaviour by opponents, the Electoral Public Prosecutor (*MPE*), or

Case Volume (2020)

Type	Volume	Inclu
Candidacy registration (RRC)	~994k	>
Coalition registration (DRAP)	~137k	>
Admin. procedural classes	various	>
Eligibility challenges		✓
Abuse of power		✓
Campaign conduct		✓
Propaganda violations		✓
Document/information offenses		✓

Adversarial filter applied at data construction stage (class and subject exclusions).

- **Restriction:** we keep only cases filed *strictly before* election day:

$$\text{filing date} < \text{election date}$$

- **Rationale:** only pre-election cases can plausibly affect competitive dynamics through legal costs, reputational signaling, or candidacy threats
- Post-election *cassação* proceedings determine whether the *realized winner* is removed — a distinct phenomenon that could reverse causality
- Applied identically to both the 2020 baseline (for shares) and the 2024 measurement (for treatment)
- **Future robustness:** *[test with >30 days before election day to address late strategic filings]*

OLS is likely biased:

1 Competition → lawsuits

Close races attract more suits; positive confound with competitiveness

2 Vulnerability → lawsuits

Incumbent weakness invites challengers to file; negative confound with incumbency

3 Unobservables

Local political culture, lawyer density, party intensity — correlated with both

Direction of bias: **ambiguous**

Identification Need

We need exogenous variation in the *level* of judicialization independent of the municipality's own 2024 competitive dynamics.

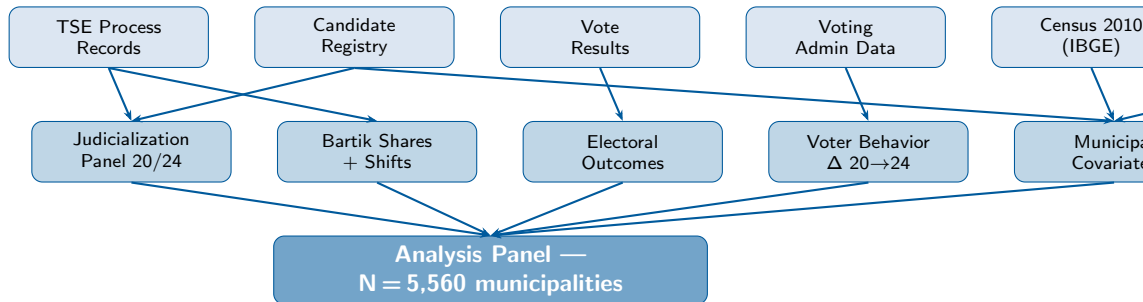
Our Strategy

Bartik shift-share IV:

Predicted 2024 lawsuits from the municipality's 2020 topic mix $\times$ state-level topic growth — a source of variation the municipality does not drive.

Theoretical framework: Helmke & Staton (2011) — litigation waves in Latin America driven by institutional and strategic diffusion at

Data Pipeline Overview



Source: TSE vote return records by zone and candidate

Processing:

- 1 Keep first-round results, mayoral races
- 2 Aggregate zones $\rightarrow$ municipality totals
- 3 Rank candidates by vote share; compute Δ 2024–2020

Second round: Only ~ 222 municipalities ($>200k$ voters) eligible. Focus on **round-1 outcomes**, decisive for 96% of municipalities.

Outcome Families

Electoral competition:

- Δ runner-up vote share
- Δ winner–runner-up margin
- Δ winner vote share
- $\Delta P(\text{winner} > 50\%)$
- Δ log number of candidates

Voter behavior:

- Δ turnout rate
- Δ null and blank vote rates

Composition:

- Δ female/nonwhite vote share
- Winner identity (new entrant vs. incumbent)

Summary Statistics

Variable	Mean	SD	Min	Max
<i>Treatment and instrument</i>				
$\Delta \log(1 + \text{adversarial lawsuits})$	-0.062	0.649	-6.2	5.1
Bartik IV (Z_m^B , adversarial filter)	-0.137	0.161	-1.3	0.4
<i>Outcome variables</i>				
Δ Runner-up vote share	-0.012	0.159	-0.500	0.499
Δ Margin (winner—runner-up)	0.060	0.325	-1.000	1.000
Δ Winner vote share	0.048	0.191	-1.000	0.718
Δ Blank vote rate	0.001	0.009	-0.071	0.070
Δ Turnout rate	0.010	0.031	-0.158	0.149
<i>Pre-determined controls</i>				
Log population (2010)	9.41	1.15	6.69	16.23
Urban share (2010)	0.080	0.144	0	1
Log income per capita (2010)	0.42	0.51	0.00	2.39
Margin 2016 (pp)	18.5	21.2	0.0	100.0
<i>N = 5,560 municipalities (baseline analysis sample)</i>				

In 2020, each municipality has a topic mix:

Some had mostly candidate eligibility challenges; others had more campaign conduct or propaganda violations.

By 2024, adversarial litigation waves diffused unevenly across topics.

Waves spread through TRE jurisprudence, coordinated party strategies, and MPE enforcement priorities — systemic forces that are state-wide, not municipality-specific.

The instrument:

Z_m^B = municipality's *exposure* to these waves, predicted from its 2020 topic mix $\times$ state-level growth in each topic.

Key Design Logic

- 1 Shares:** from 2020 — pre-determined, before 2024 politics materialize
- 2 Shifts:** state-level waves driven by institutional and strategic diffusion, not by any single municipality's dynamics
- 3 Product:** predicted exposure to adversarial litigation waves, orthogonal to local 2024 competitive outcomes

Template

Ash, Morelli & Vannoni (2025, JPE): Bartik IV for legislative output, $\times$ state growth

$$Z_m^B = \sum_{k \notin \mathcal{A}} s_{m,k,2020} \cdot g_{UF(m), k, -m}$$

$s_{m,k,2020}$	Municipality m 's share of topic k among adversarial lawsuits in 2020; re-normalized so that $\sum_k s_{m,k,2020} = 1$ per municipality
$g_{UF,k,-m}$	Leave-one-out log growth of topic k in state $UF(m)$, 2020 to 2024: $g = \log\left(\frac{T_{UF,k,2024} - C_{m,k,2024}}{T_{UF,k,2020} - C_{m,k,2020}}\right)$
$\mathcal{A}$	Set of excluded administrative/procedural classes and subjects (incl. RRC, DRAP, mandatory class codes)
LOO	Municipality m excluded from its own state shift to prevent mechanical correlation

Single preferred specification. The adversarial filter is applied *at the data construction stage* via class- and subject-code exclusions, not post-hoc topic removal. This produces one instrument ($F \approx 19.6$) used throughout.

Formal condition:

$$E[Z_m^B \cdot u_m \mid \mathbf{X}_m, \delta_{UF}] = 0$$

GPS (2020) identification approach:

Validity rests on **conditional exogeneity of the 2020 shares** $s_{m,k,2020}$. A municipality's 2020 litigation portfolio was determined by then-prevailing political dynamics — two election cycles before 2024 outcomes materialize.

BHJ (2022) shift exogeneity:

State-level topic growth reflects institutional and strategic waves (TRE jurisprudence, MPE priorities, party legal strategies) that are common to a state and not driven by individual municipality dynamics.

Potential Threats

- 1 Large-municipality dominance:** m drives state shift
→ LOO construction; most municipalities are small
- 2 Share persistence:** 2020 shares correlated with unobserved 2024 dynamics
→ state FE + 2020 competitive controls; balance table (Appendix F)
- 3 Network spillovers:** litigation waves correlate with state political events
→ state FE absorbs common state shocks

Which litigation topics actually drive the instrument?

The 2SLS estimand decomposes as $\hat{\tau}_{IV} = \sum_k \alpha_k \hat{\tau}_k$, where $\alpha_k = (\tilde{z}'_k \tilde{d}) / (\tilde{Z}' \tilde{d})$ is topic k 's share of the first-stage numerator and $\hat{\tau}_k$ is the just-identified IV using only topic k .

Rank	Topic (abbreviated)	α (%)	Cum.	F_k	$\hat{\tau}_k^{\text{majority}}$
1	Document falsification (electoral)	+23.5	0.24	50.9	-0.138
2	Fraudulent electoral poll	-22.6	0.46	60.0	—
3	Right of reply	+16.4	0.62	1.1	-3.35
4	Election general (1st round)	+15.8	0.78	1.9	-0.325
5	Multiple party registration	-15.4	0.93	36.5	—
Top 5 (of ~30 adversarial topics)		+93.7			
HHI = 0.34; $K_{\text{eff}} = 1/\text{HHI} \approx 3.0$. Topics 3–4 have high α_k but near-zero F_k : GPS concern.					

GPS concern: high α , low F_k

Topics 3–4 (Right of Reply, Election 1st Round) have large Rotemberg weights but $F_k < 2$. These topics contribute to the first-stage numerator but carry little independent identifying variation. Visual IV graph (Appendix E) assesses sign consistency across all topics.

Well-identified topics

Topics 1 (doc. falsification, $F_k = 50.9$) and 2 (fraudulent poll, $F_k = 60.0$) have both high $|\alpha_k|$ and strong per-topic first stages — the cleanest contribution to identification.

GPS/BHJ Validity Checklist

Requirement	Status	Key result
Report Rotemberg weights α_k	✓ Done	HHI = 0.34; $K_{\text{eff}} \approx 3$
Per-topic first-stage F_k	✓ Done	F_k ranges 1–60 across topics
Visual IV graph (τ_k vs F_k)	✓ Done	Appendix E; sign consistency checked
Balance: Z_m^B on pre-determined chars	✓ Done	3/9 covariates $p < 0.10$; all in controls
Leave-one-out shifts (LOO)	✓ Built in	Prevents mechanical correlation
Exposure-robust (AKM) SEs	✓ Done	SE ratio $\approx 2\text{--}3\times$; $K_{\text{eff}} = 3$
Shift descriptives table (BHJ)	✓ Done	See <code>shift_descriptives.csv</code>
tF correction (Lee et al. 2022)	✓ Done	$F = 19.6 < 23.1$; $tF_{\text{cv}} = 2.18$
LIML comparison (GPS robustness)	✓ Done	LIML = 2SLS (exact, $K=1$)
Topic share controls (AMV/BHJ)	✓ Done	Spec robustness topic shares

First stage (OLS):

$$\underbrace{\Delta \log(1 + \ell_m)}_{\text{endogenous}} = \alpha + \beta Z_m^B + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + \varepsilon_m$$

Second stage (2SLS):

$$\Delta Y_m = \alpha + \tau \Delta \widehat{\log(1 + \ell_m)} + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + u_m$$

- ℓ_m = adversarial competition-relevant pre-election lawsuits (respondent is mayoral candidate)
- Δ = first difference 2024–2020
- $\mathbf{X}_m$ = 7 baseline pre-determined controls
- $\delta_{\text{UF}(m)}$ = state fixed effects (27 units)
- SE clustered by principal electoral zone (2,187 clusters)
- tF critical values (Lee et al. 2022) reported for weak-instrument-robust inference given $F = 19.6 < 23.1$

Eight Specifications

Specification	Controls	FE	Sample	N
baseline_state_fe	7 baseline	State (27)	All	5,560
baseline_state_fe_sz	7 baseline	State (27)	Single-zone	5,371
robustness_full_controls	14 extended	State (27)	All	5,560
subsample_le200k	7 baseline	State (27)	$\leq 200k$ voters	5,338
subsample_open_seat	7 baseline	State (27)	Open seat (46%)	2,571
subsample_contested_seat	7 baseline	State (27)	Contested (54%)	2,989
robustness_topic_shares	7 + 4 topic shares	State (27)	Nonzero shares	2,824
robustness_broader_lawsuits	7 + log(lawsuits no-RRC 2020)	State (27)	All	5,560

7 baseline controls: Census 2010 (log pop, urban share, log income pc) + margin 2016 + 2020 electoral baseline (log valid votes, margin top1–top2, log candidates)

14 extended: baseline + ENP 2016 + candidate composition 2020 (gender, ethnicity, education) + party count

Open-seat definition: a municipality is *open seat* in 2024 if the 2020 winner also won in 2016 (two consecutive terms), making them constitutionally term-limited. This is determined by 2016–2020 outcomes, predating 2024 judicial activity. **Note on topic-shares spec:** F collapses to <1 (mechanical; shares ARE the instrument's

First-Stage Results

Specification	Coef.	SE	t	F-stat	tF_{cv}	N
Baseline (state FE)	1.791	0.405	4.43	19.6	2.18	5,560
Baseline, single-zone	1.801	0.422	4.27	18.2	2.23	5,371
Full controls	1.799	0.406	4.44	19.7	2.18	5,560
$\leq 200k$ voters	1.869	0.426	4.39	19.3	2.19	5,338
Broader lawsuits control	1.792	0.405	4.43	19.6	2.18	5,560
Topic shares control*	-0.656	0.926	-0.71	0.5	—	2,824

SE clustered by principal electoral zone. $F = (t\text{-stat})^2$. tF_{cv} from Lee et al. (2022) Table 1.

*Topic-shares spec: F collapses because the 2020 shares ARE the instrument's variation (expected by construction).

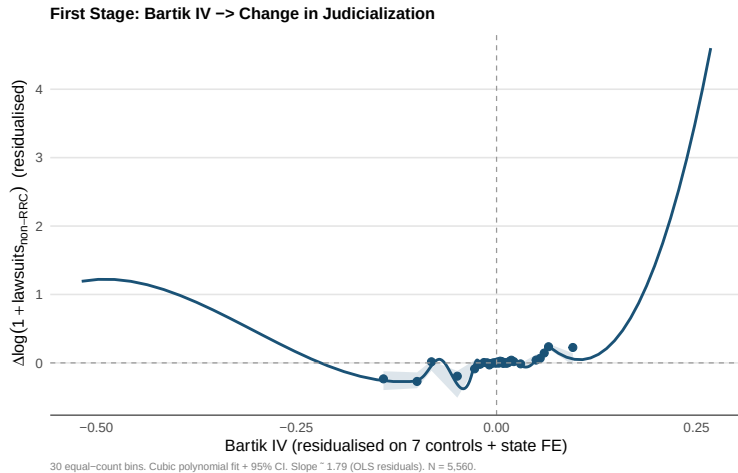
Instrument Strength

$F = 19.6$ across all specifications excluding the topic-shares sensitivity (where collapse is mechanical). Stock-Yogo 10% size distortion critical value ≈ 10 ; tF threshold for 5% inference = 23.1. We apply tF correction throughout.

LIML Check

For a just-identified model ($K = 1$ instrument), $LIML \equiv 2SLS$ exactly. Confirmed: max coefficient divergence = 0.0000 across all outcomes. No many-instrument or weak-instrument bias.

First Stage — Binscatter



Electoral Competition — Baseline Specification

Adversarial instrument ($F = 19.6$). $N = 5,560$, clusters = 2,187. State FE + 7 controls.

Outcome	Coef.	SE	p	t	tF reject?
Δ Runner-up vote share	-0.031	0.022	0.157	-1.42	×
Δ Margin (W-RU)	+0.062	0.043	0.151	+1.44	×
Δ Winner vote share	+0.031	0.025	0.213	+1.25	×
Δ Winner majority (P50)	+0.140	0.090	0.122	+1.55	×
Δ Log n candidates	-0.066	0.039	0.092	-1.69	×

Directional pattern (winner consolidation)

All competition outcomes point toward winner consolidation: higher majority probability, wider margin, fewer candidates. Direction is consistent across all 6 specifications. **None survives tF correction** ($tF_{cv} = 2.18$; $\max |t| = 1.69$).

Candidate pool: marginal signal

Log candidates (-6.6% , $p = 0.092$) is the only outcome approaching conventional significance. Consistent with Besley & Coate (1997): adversarial challenges raise the fixed cost of candidacy, deterring entry at the margin. Replicated in legislative analysis (-7.5% , $p = 0.066$).

Electoral Competition — Robustness Across Specifications

Outcome: Δ winner majority. All specifications, adversarial instrument.

Specification	Coef.	SE	p	F	N
Baseline (state FE)	+0.140	0.090	0.122	19.6	5,560
Baseline, single-zone	+0.131	0.093	0.160	18.2	5,371
Full controls	+0.136	0.086	0.116	19.7	5,560
$\leq 200k$ voters	+0.128	0.091	0.161	19.3	5,338
Broader lawsuits control	+0.140	0.090	0.122	19.6	5,560

- Direction and magnitude are remarkably stable: coefficient ranges +0.128 to +0.140 across all specifications
- The result is not driven by control set, sample, or the non-adversarial baseline
- The precision gap (not reaching $p < 0.05$) is the fundamental limitation; a larger panel or stronger instrument would be needed to confirm
- *[Pre-period placebo (2016→2020 outcomes): required to rule out pre-existing trend confounds]*

Voter Behavior — Baseline Specification

$N = 5,560$, state FE + 7 controls. Rates denominated on registered voters.

Outcome	Coef.	SE	p	t	tF reject?
Δ Turnout rate	-0.006	0.005	0.185	-1.33	×
Δ Null vote rate	+0.007	0.006	0.220	+1.23	×
Δ Blank vote rate	+0.013	0.005	0.018	+2.36	✓
Δ Valid vote rate	-0.026	0.013	0.038	-2.07	×

Main finding: blank vote rate

Adversarial judicialization $\uparrow$ by 1 SD (≈ 0.16 units) $\Rightarrow$ blank rate $\uparrow$ by $0.013 \times 0.16 = 0.2$ pp. At the mean blank rate of 1.9%, this is a $\approx 10\%$ increase.

Effect survives tF correction ($|t| = 2.36 > tF_{cv} = 2.18$).

Valid-vote rate ($p = 0.038$) is the accounting mirror and does *not* survive tF. **Blank rate is the primary identified effect.**

Precise nulls

Turnout ($p = 0.185$) and null rate ($p = 0.220$) are precise nulls. Adversarial judicial challenges do *not* demobilize the electorate, contrary to negative advertising effects (Ansolabehere & Iyengar 1995). The behavioral channel runs through expressive abstention, not non-participation.

What blank votes signal:

In Brazil, a blank vote (*voto em branco*) is a deliberate choice to reject all listed candidates without spoiling the ballot. It requires specific action (pressing a button or writing 'branco').

Mechanism:

Adversarial judicial challenges are publicly filed records that generate information about candidates' alleged conduct. Voters exposed to more diverse and intense judicial activity may face:

- Increased uncertainty about all candidates' quality
- Negative updating about the entire political class (*not* specific to a targeted candidate)
- Expressive dissatisfaction with the judicially-contested electoral environment

Literature anchor

Ferraz & Finan (2008): publicly released audit information changes electoral outcomes in Brazil. Judicial filings are analogous: they are publicly available before election day and carry information about alleged wrongdoing.

Key difference: audits target specific corrupt incumbents (directional information); adversarial judicial waves are diffuse across all candidates (ambiguity-generating). This explains the blank vote outcome rather than an incumbent vote share effect.

Robustness

Effect stable: coefficient 0.012–0.013 across baseline, single-zone, and full-controls specifications.

Open seat (46% of municipalities, $N = 2,571$):
2020 winner served two consecutive terms $\Rightarrow$
constitutionally term-limited in 2024. No
incumbent on the ballot. Exogenous to 2024
litigation activity.

Contested seat (54%, $N = 2,989$):
2020 winner's first term. Incumbent can seek
reelection. Voters have a focal candidate
(support or oppose the incumbent).

Sample	Coef.	SE	p	F	tF?
All	+0.013	0.005	0.018	19.6	✓
Open seat	+ 0.030	0.014	0.028	10.6	✗*
Contested	+0.003	0.005	0.546	15.3	✗

* Conventional $p = 0.028$ but $|t| = 2.20 < tF_{cv} = 2.78$ ($F = 10.6$).

Mechanism

The aggregate blank-rate effect runs almost entirely through **open-seat municipalities**.

Why? When no incumbent is running, voters lack a focal candidate. Adversarial judicial activity creates ambiguity about *all* contenders simultaneously — heightening the attractiveness of a blank vote.

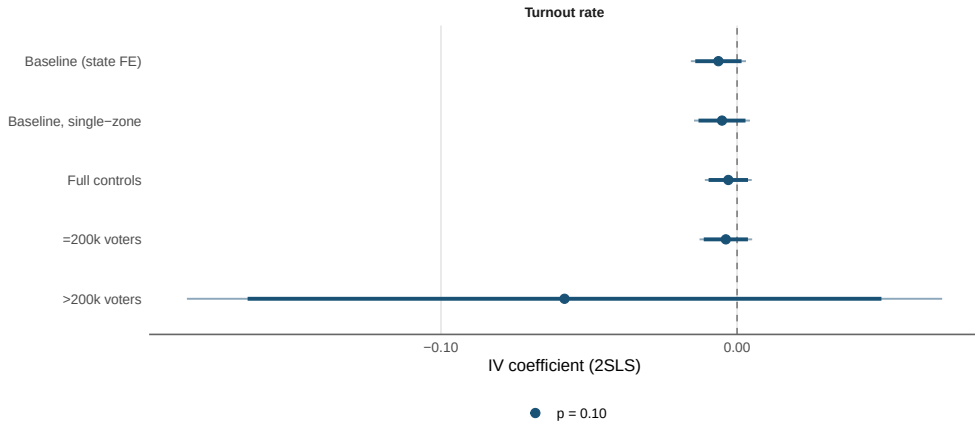
When the incumbent runs, voters already have a valence benchmark (vote for or against the mayor). Legal noise is absorbed: the contested-seat effect is a structural zero.

Caveat: weakened instrument

Open-seat subgroup: $F = 10.6$ (vs. 19.6 full sample). Smaller municipalities likely to have longer-serving incumbents but also less electoral litigation variation. The tF correction is therefore stricter ($tF_{cv} = 2.78$); effect survives conventional but not weak-IV-robust inference.

Voter Behavior — Forest Plot (All Specs)

Voter Behavior Outcomes — All Specifications



Thick bars: 90% CI. Thin bars: 95% CI. SE clustered by principal electoral zone.

Candidate Pool and Composition — Baseline

Adversarial instrument, baseline specification. $N = 5,560$, clusters = 2,187.

Outcome	Coef.	SE	p	tF reject?
Δ Log n candidates (exec.)	-0.066	0.039	0.092	×
Δ New-cand. vote share (exec.)	-0.010	0.050	0.848	
Δ Incumbent vote share (exec.)	+0.010	0.049	0.844	
Δ Female vote share	-0.093	0.048	0.052	×
Δ Winner is female	-0.123	0.063	0.052	×
Δ Winner is new entrant	-0.002	0.073	0.982	

Entry deterrence (not composition shift)

The candidate count reduction (-6.6%) is directional, not significant. But it replicates across executive and legislative arenas (next slide), lending it more credibility. New-entrant and incumbent vote shares are both near zero — the field shrinks, but without systematic reallocation toward any type.

Gender: directional, not robust

Female vote share (-9.3 pp, $p = 0.052$) and female winner probability (-12.3 pp, $p = 0.052$) are just below 5% but do not survive tF correction ($|t| = 1.94$ and $1.94 < 2.18$). Direction is suggestive — adversarial legal challenges may disadvantage women candidates — but requires richer candidate-level data to confirm.

Three entrant types, 4 electoral cycles (2012–2024):

- **Pure new entrant:** no prior mayoral candidacy anywhere in 2012, 2016, or 2020 ($\bar{s}_{2024} = 55\%$)
- **Serial challenger:** ran in the same municipality in 2020, lost, running again ($\bar{s}_{2024} = 13\%$)
- **Cross-cycle returner:** has prior history but sat out 2020 entirely; returning after a cycle gap ($\bar{s}_{2024} = 9\%$)

Question: Does adversarial judicialization reshape *who enters*, not just *how many*?

IV Results: Entry Composition

Outcome	Coef.	SE	<i>p</i>
Δ New entrant share	−0.011	0.043	0.793
Δ Serial challenger	+0.007	0.026	0.802
Δ Cross-cycle ret.	+0.005	0.025	0.861
Baseline spec. N = 5,560, F = 19.6.			

Interpretation

All three entrant-type shares are precise nulls. Judicialization does not *selectively screen* entrant types — it reduces the count (−6.6%) without reshaping the pool's experience composition.

Entry deterrence is a **quantity** effect, not a **selection** effect.

Same instrument, different electoral arena:

- Brazil's city councils (*Câmaras Municipais*) elect 9–55 *vereadores* depending on population (CF Art. 29-IV)
- Pre-election challenges also target legislative candidates
- **Multi-seat proportional** system — different from the plurality mayoral race

Identification: same municipality-level Bartik IV ($F \approx 19.4$). The shift in adversarial litigation waves applies to the same municipality and electoral environment. Endogenous: $\Delta \log(1 + \text{competition lawsuits})$ at municipality level (same treatment).

Outcome Families

Candidate pool:

- Total candidates, female/nonwhite shares
- New candidates, incumbents, party count

Elected composition:

- Elected female/nonwhite shares
- Higher-education share, mean age
- Incumbent reelected share

Party competition:

- Party count, coalition count

Instrument

$N = 5,560$; clusters = 2,187; $F = 19.4$.

Baseline specification. N = 5,560, clusters = 2,187. State FE + 7 controls. F = 19.4.

Outcome	Coef.	SE	p	Sig.
<i>Candidate pool</i>				
Δ Log total candidates	−0.075	0.041	0.066	†
Δ Female candidate share	−0.006	0.006	0.370	
Δ Nonwhite candidate share	−0.018	0.019	0.346	
Δ New candidate share	−0.020	0.016	0.217	
Δ Eff. party count (candidates)	−0.264	0.301	0.381	
<i>Elected composition</i>				
Δ Elected female share	+0.013	0.022	0.554	
Δ Elected nonwhite share	−0.027	0.026	0.288	
Δ Incumbent reelected share	+0.006	0.025	0.797	
<i>Party competition</i>				
Δ Party count	−0.260	0.354	0.463	
Δ Coalition count	+0.074	0.066	0.266	

† $p < 0.10$. SE clustered by principal electoral zone.

Same instrument ($F \approx 19\text{--}20$), two electoral arenas:

Outcome	Executive	Legislative
Δ Log candidates	$-0.066^\dagger$	$-0.075^\dagger$
Winner consolidation	directional	—
Female share	$-0.093^\dagger$	-0.006
Blank rate	$+0.013^{**}$	—
Elected composition	null	null
Party competition	null	null

$**p < 0.05$ (tF corrected); $^\dagger p < 0.10$ (uncorrected).

Interpretation

Both arenas: candidate pool contracts (-7%) consistently — entry deterrence channel is the most robust structural finding. Consistent with Besley & Coate (1997).

Executive only: winner consolidation (directional), blank rate (significant), gender effects (borderline) — these run through competition dynamics specific to the single-winner plurality race.

Legislative: multi-seat proportional allocation dampens the competitive displacement effects visible in executive races. No effect on who wins seats or on party composition.

Motivation:

The aggregate Bartik IV combines topics from four litigation families. Do different *types* of adversarial challenges operate through different mechanisms?

Approach (AMV 2025, Section 6):

Construct a *family-specific* Bartik IV for each topic family:

$$Z_m^f = \sum_{k \in f} s_{m,k,2020} \cdot g_{UF,k,-m}$$

Each Z_m^f instruments the *family-specific* endogenous: $\Delta \log(1 + \ell_{m,f})$. This identifies a family-level LATE — different from the aggregate.

Four Litigation Families

Family	F	N	Identified?
Campaign conduct	36.0	3,937	✓
Information env.	19.7	3,861	✓
Abuse of office	7.8	3,311	✗
Eligibility/Ballot	8.4	1,772	✗

Campaign conduct: campaign propaganda, prohibited conduct, vote-buying

Information environment: media, internet propaganda, right of reply

Eligibility/Ballot access: registration challenges, impugnação

Abuse of office: economic and political power abuses

Family IV Results — Primary Outcomes

Family-specific Bartik IV instruments family-specific endogenous. Baseline state FE + 7 controls.

Family (F^{fs})	Δ Runner-up		Δ Margin		Δ Log cand.	
	Coef.	p	Coef.	p	Coef.	p
Campaign ($F = 36$)	-0.007	0.545	+0.018	0.442	-0.004	0.836
Info. env. ($F = 20$)	-0.010	0.511	+0.034	0.250	+0.013	0.665
Abuse of office ($F = 7.8$) and eligibility ($F = 8.4$): weak instrument — estimates unreliable.						

Null across well-identified families

Both campaign-conduct and information-environment sub-instruments produce null effects across all competition outcomes. This is consistent with the aggregate null — the aggregate result is not concealing strong but cancelling effects across family types. Direction is non-systematic (not all positive or all negative), further supporting the null.

Implications

The blank-rate effect and the candidate-pool reduction are not attributable to a single mechanism family with the currently available family instruments. Both campaign conduct and information environment challenges contribute to the aggregate treatment without producing detectable competition effects through their separate channels.

Art. 29-II CF/88: municipalities with $>200,000$ registered voters hold a mandatory second round if no candidate wins $>50\%$

Two subsamples:

- $\leq 200k$ voters: 5,338 — strictly single-round; homogeneous institutional regime
- $>200k$ voters: 222 — conditional second round; instrument loses power ($F < 1$)

Large-city failure: consistent with instrument design — larger municipalities tend to dominate their state shifts, so the LOO construction leaves less identifying variation.

Δ Winner majority by subsample

Sample	Coef.	SE	p	
All ($N = 5,560$)	+0.140	0.090	0.122	19
$\leq 200k$ ($N = 5,338$)	+0.128	0.091	0.161	19
$>200k$ ($N = 222$)	n.a.			4

Single-zone municipalities

$N = 5,371$ (96.6%): municipality = zone = cluster; exact correspondence. Coefficients essentially identical to full sample; $F = 18.2$.

14-variable extended control set:

Adds ENP 2016, candidate composition (gender, ethnicity, education), party/coalition count, voter behavior baseline, candidate experience.

Caveat: some extended controls are outcome-adjacent (2020 candidate composition may reflect prior judicialization). Use as sensitivity only.

Broader lawsuits control:

Adds $\log(1 + \ell_m^{\text{no-RRC}})$ in 2020 as a covariate — the broader competition lawsuit stock (excluding only RRC). Tests whether the adversarial IV effect is distinct from general pre-existing litigation activity.

Blank rate across control sets

Spec	Coef.	SE	<i>p</i>	F
7 baseline	+0.013	0.005	0.018	19.6
14 extended	+0.012	0.005	0.022	19.7
Broader ctrl	+0.013	0.005	0.018	19.6
Margin (7 base.)	+0.062	0.043	0.151	19.6
Margin (14 ext.)	+0.057	0.042	0.180	19.7
Margin (broader)	+0.062	0.043	0.151	19.6

Blank rate result is stable to control set and to conditioning on broader lawsuit baseline.

1 Blank vote rate — main finding:

+1.3 pp ($p = 0.018$; survives tF correction).

Adversarial judicialization generates voter ambivalence, expressed through deliberate blank votes. Turnout and null rate: precise nulls. No demobilization.

2 Candidate pool contraction (marginal):

−6.6% executive ($p = 0.092$); −7.5% legislative ($p = 0.066$). Consistent across arenas — entry deterrence is the most robust structural channel.

3 Winner consolidation (directional):

All competition outcomes favor the winner: higher majority, wider margin. Direction stable across all 5 core specifications. Not statistically identified after tF correction.

Main Takeaway

Credibly identified estimates from a Bartik shift-share IV ($F = 19.6$, tF-corrected inference).

The primary causal effect of adversarial judicialization on Brazilian municipal elections is voter ambivalence — a significant rise in blank votes — rather than competitive displacement or mobilization.

Entry deterrence is the most consistent structural finding across arenas. Winner consolidation is the most economically plausible pattern; it remains statistically underpowered.

Contributions:

- First Bartik IV applied to electoral judicialization; blueprint for causal identification of litigation effects in political economy
- Full GPS/BHJ/AKM compliance; tF correction applied
- Systematic administrative-vs-adversarial distinction: a measurement contribution to the electoral law literature
- Cross-arena comparison (executive vs. legislative): entry deterrence replicates; competition effects arena-specific

Open questions requiring new data:

- *[Pre-period placebo: 2016→2020 outcomes as falsification test]*
- *[Case outcomes (win/loss rulings) as second instrument]*

Pending analysis

- Romano–Wolf multiple testing correction across outcome families
- Concavity analysis: zero-share vs. positive-share municipalities (intensive vs. extensive margin)
- ✓ Open-seat heterogeneity: done (blank rate $3\times$ larger in open seats)
- ✓ Entry typology: done (quantity effect, not selection)
- Female candidate heterogeneity: candidate-level analysis of gender differential exposure

Literature Connections

GPS (2020); BHJ (2022); AKM (2019); Lee et

Appendix A: Full First-Stage Table I

Adversarial instrument (bartik_iv_2020_2024). Single specification.

Adversarial filter applied at data construction stage (class and subject-code exclusions from TSE taxonomy; mandatory administrative filings excluded). One instrument, one endogenous variable.

Specification	Coef.	SE	<i>t</i>	F-stat	tF _{cv}	<i>N</i>
Baseline (state FE)	1.791	0.405	4.43	19.6	2.18	5,560
Baseline, single-zone	1.801	0.422	4.27	18.2	2.23	5,371
Full controls	1.799	0.406	4.44	19.7	2.18	5,560
≤200k voters	1.869	0.426	4.39	19.3	2.19	5,338
Broader lawsuits ctrl	1.792	0.405	4.43	19.6	2.18	5,560
Topic shares ctrl*	−0.656	0.926	−0.71	0.5	—	2,824

SE clustered by principal electoral zone. $F = (t\text{-stat})^2$.

tF_{cv}: weak-IV-robust critical value (Lee et al. 2022, Table 1), 5% level.

*Identification collapses: the 2020 shares ARE the instrument's variation.

Appendix B: Full IV Table — Competition and Voter Behavior I

Adversarial instrument. Baseline specification. N = 5,560, clusters = 2,187.

Outcome	Coef.	SE	<i>t</i>	<i>p</i>	tF*?
<i>Competition</i>					
Δ Runner-up	−0.031	0.022	−1.42	0.157	×
Δ Margin	+0.062	0.043	+1.44	0.151	×
Δ Winner vote	+0.031	0.025	+1.25	0.213	×
Δ Majority	+0.140	0.090	+1.55	0.122	×
Δ Log candidates	−0.066	0.039	−1.69	0.092	×
<i>Voter behavior</i>					
Δ Turnout rate	−0.006	0.005	−1.33	0.185	×
Δ Null rate	+0.007	0.006	+1.23	0.220	×
Δ Blank rate	+0.013	0.005	+2.36	0.018	✓
Δ Valid vote rate	−0.026	0.013	−2.07	0.038	×

*tF_{cv} = 2.18 (F = 19.6); ✓ = |*t*| > 2.18. SE clustered by principal electoral zone.

Adversarial instrument. Baseline state FE. N = 5,560.

Outcome	Coef.	SE	<i>t</i>	<i>p</i>
Δ Female vote share	−0.093	0.048	−1.94	0.052
Δ Nonwhite vote share	−0.039	0.042	−0.94	0.348
Δ New cand. vote share	−0.010	0.050	−0.19	0.848
Δ Incumbent vote share	+0.010	0.049	+0.20	0.844
Δ Winner is female	−0.123	0.063	−1.94	0.052
Δ Winner is new entrant	−0.002	0.073	−0.02	0.982
Δ Winner ideology	−0.168	0.205	−0.82	0.412
Pedersen volatility	−0.010	0.038	−0.25	0.801

SE clustered by principal electoral zone. $tF_{cv} = 2.18$; no composition outcome survives tF correction.

Competition-relevant adversarial case criteria:

- 1 Mayoral candidate (*prefeito*) is the respondent
- 2 Filed before election day
- 3 Case class and subject are not on the administrative exclusion list

Examples of included subjects:

Abuso do Poder Econômico, Abuso do Poder Político, Captação Ilícita de Sufrágio, Propaganda Eleitoral Irregular, Conduta Vedada a Agente Público, Uso Indevido dos Meios de Comunicação, Impugnação de Registro de Candidatura, Falsificação de Documento

Case Volume (2020 / 2024)

Category	2020	2024
Mandatory registration (RRC)	~994k	~841k
Coalition registration (DRAP)	~137k	~118k
Adversarial competition cases	~385k	~340k

Filter logic

The adversarial filter excludes entire administrative *classes* (mandatory processing) as well as specific *subjects* that are procedurally rather than substantively adversarial. This is stricter than simply removing RRC and DRAP topic entries.

Figure E1: Distribution of Bartik IV

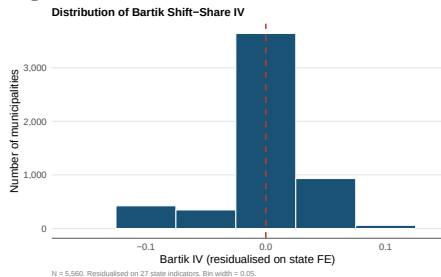
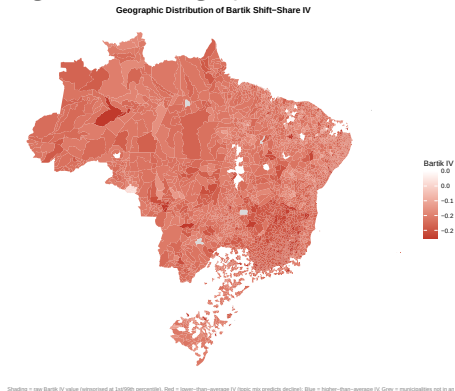


Figure E2: Geographic distribution



IV residualized on state FE for histogram. Map: raw Bartik IV by municipality (grey = not in sample). Visual IV graph (τ_k vs. F_k , sized by $|\alpha_k|$): see `output/figures/visual_iv_grid.pdf`.

Appendix F: Balance Table I

Adversarial instrument regressed on each covariate. State FE, SE clustered by zone.

Covariate	Coef.	SE	p
Log population (2010)	-0.543	0.256	0.034
Urban share (2010)	-0.024	0.028	0.385
Log income per capita (2010)	+0.164	0.049	0.001
Margin 2016	+0.089	0.041	0.030
ENP 2016	-0.256	0.142	0.072
Turnout rate 2020	+0.040	0.012	0.001
Female candidate share 2020	-0.075	0.046	0.101
Nonwhite candidate share 2020	-0.050	0.067	0.458
Log n candidates 2020	-0.127	0.084	0.129

Interpretation: Non-random

3 of 9 covariates $p < 0.05$. Correlations reflect that larger, richer, higher-turnout municipalities had more diverse 2020 litigation portfolios.

All three pre-determined controls included in every specification.

share exposure is expected when municipalities differ systematically. The identification claim is

that these differences are absorbed by the pre-determined controls and state FE. The GPS balance tests (`gps_balance_tests.csv`) run more detailed regressions per top topic.

Appendix G: Family IV — First Stage by Family

Each family IV instruments its family-specific endogenous ($\Delta \log(1 + \ell_{m,f})$). Baseline spec.

Family	Endog. coef.	SE	F	N	Identified?
Campaign conduct	+1.61	0.27	36.0	3,937	✓
Information env.	+2.29	0.52	19.7	3,861	✓
Abuse of office	−2.54	0.91	7.8	3,311	✗
Eligibility/Ballot	−0.20	0.07	8.4	1,772	✗

Notes: N varies by

family because family-IV is defined only for municipalities with nonzero family exposure ($s_{mk,2020} > 0$ for topic k in family f). Negative coefficients for abuse and eligibility families indicate the Bartik IV for those families is negatively associated with own-family lawsuit growth — consistent with diffuse spillover dynamics. These families should not be interpreted.

Abuse of office and **eligibility/ballot** families have $F < 10$; weak-instrument concerns preclude reliable 2SLS inference. **Campaign conduct** ($F = 36$) and **information environment** ($F = 20$) are the identified families.

All main results estimated in **R** using
`fixest::feols()` v0.14.1:

IV formula (adversarial instrument):

```
feols(outcome ~ controls | SG_UF |  
      delta_log1p_competition_lawsuits  
      ~ bartik_iv_2020_2024,  
      data, cluster = ~cluster_id)
```

LIML (confirmation only):

```
feols(..., estimator = "liml")  
# Result: identical to 2SLS (K=1)
```

Key output files

Outcome	2SLS	LIML
Δ Majority	+0.140	+0.140
Δ Blank rate	+0.013	+0.013
Δ Log cand.	-0.066	-0.066
Max diff = 0.0000 (K=1 exact)		

Benchmark

Ash et al. (2025) Stata:
`ivreghdfe y (endog = instr),
absorb(state) cl(zone)`

Art. 29-II, Constituição Federal 1988:

“eleição do Prefeito e do Vice-Prefeito realizada no dia 1.^o de outubro do ano anterior ao do término do mandato dos que devam suceder, aplicadas as regras do art. 77, no caso de Municípios com mais de duzentos mil eleitores.”

Implication: the 200,000 registered voter threshold

determines whether a municipality follows plurality rule (one round) or the two-round runoff. This creates a sharp institutional discontinuity motivating our population subsample.

Sample Split (2024 Registered Voters)

	<i>N</i>	Share
$\leq 200,000$	5,338	96.0%
$> 200,000$	222	4.0%
Total	5,560	100%

Instrument failure in large cities

Large municipalities ($>200k$) have $F < 1$ in both the main and subsample specifications — consistent with near-violation of the LOO assumption when m is itself a large share of its state's litigation pool.